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| Assignment 1 | Project Summary |
| Course | Computer Vision - 2026 |
| GitHub Repository | https://github.com/nigosto/natural-language-video-search |

| Project author | | |
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| Project name | HPC-Accelerated Natural-Language Search of Surveillance Video |

## 1. Short project description (Business needs and system features)

The project will develop a system that allows a user to search a collection of surveillance videos using natural-language descriptions of events. Instead of manually watching many hours of footage, a user could enter a query such as “a person enters the room carrying a bag” and receive a ranked list of video segments that may match the description. Each result will include the source video, approximate start and end timestamps, and a preview.

The system will use pretrained vision-language models to estimate the similarity between a text query and sampled video content. It will compare two approaches: X-CLIP, which processes video clips, and CLIP, which represents individual frames and combines their scores over time.

The objectives are to:

1. Prepare a manageable subset of a surveillance video dataset for search and evaluation.
2. Implement text-to-video retrieval using pretrained X-CLIP and a frame-based CLIP baseline.
3. Retrieve and rank candidate segments for a user’s natural-language query.
4. Combine nearby matching samples into timestamped segments and display the results.
5. Evaluate retrieval quality against annotated event descriptions and timestamps.
6. Measure the effects of model choice, frame-sampling rate, batch size, and CPU/GPU processing on runtime and retrieval quality.
7. Provide a user interface for submitting queries and reviewing results.

## 2. Methodology

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| **Dataset preparation** | Inspect the data format, remove unusable files, and split the annotated examples into development and evaluation sets where the dataset size permits. The same videos and queries will be used to compare the two models. |

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| **Video preprocessing** | Decode each video and sample frames or short clips at a configurable interval. Resize and normalize the sampled content according to each model's input requirements. Record the timestamp associated with every sample so that retrieved matches can be mapped back to the original video. |
| **Model Components** | • **CLIP baseline:** Encode sampled frames and the text query using a pretrained CLIP checkpoint. Calculate frame-to-text similarity scores, then combine nearby frame scores to estimate candidate segment relevance.<br>• **X-CLIP:** Encode sampled video clips and text queries using a pretrained X-CLIP checkpoint. Calculate similarity scores between the query and each clip representation. |
| **Segment generation and ranking** | Rank samples by text-video similarity. Apply temporal post-processing to group nearby high-scoring samples into candidate intervals. Parameters such as the similarity threshold, allowed gap between matches, and minimum segment duration will be selected using the development set, then held fixed for evaluation. Return the top-ranked intervals with timestamps and scores. |
| **Evaluation** | Measure retrieval quality and runtime for both models. Run controlled experiments that vary selected parameters, such as sampling interval, inference batch size, and CPU versus GPU execution. Report the trade-offs between retrieval quality, processing cost, and query latency. |

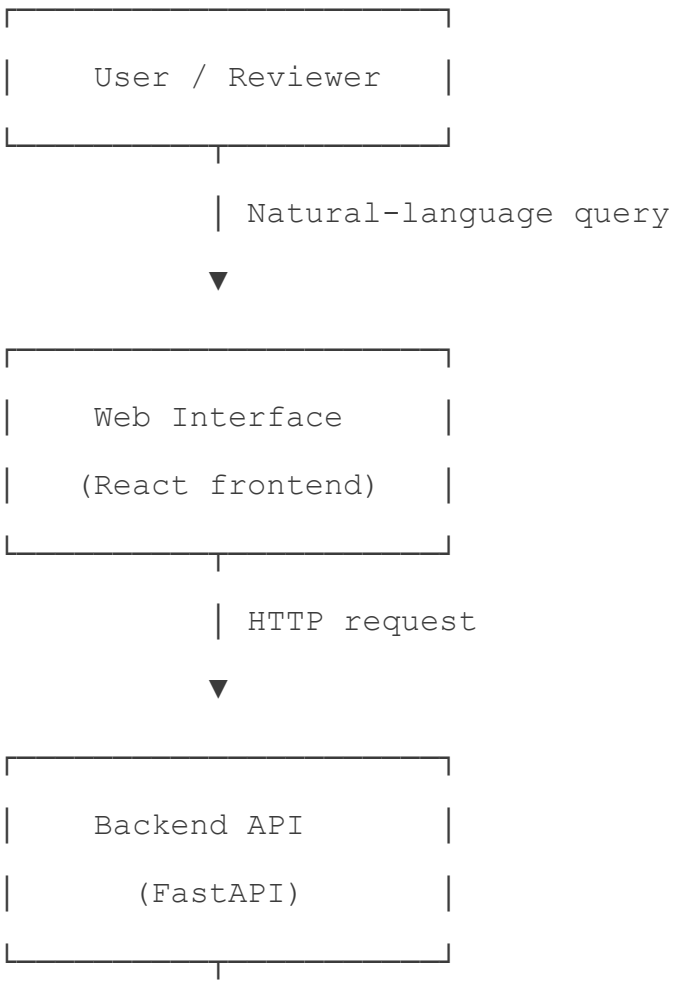
## 3. Main Use Cases / Usage Scenarios

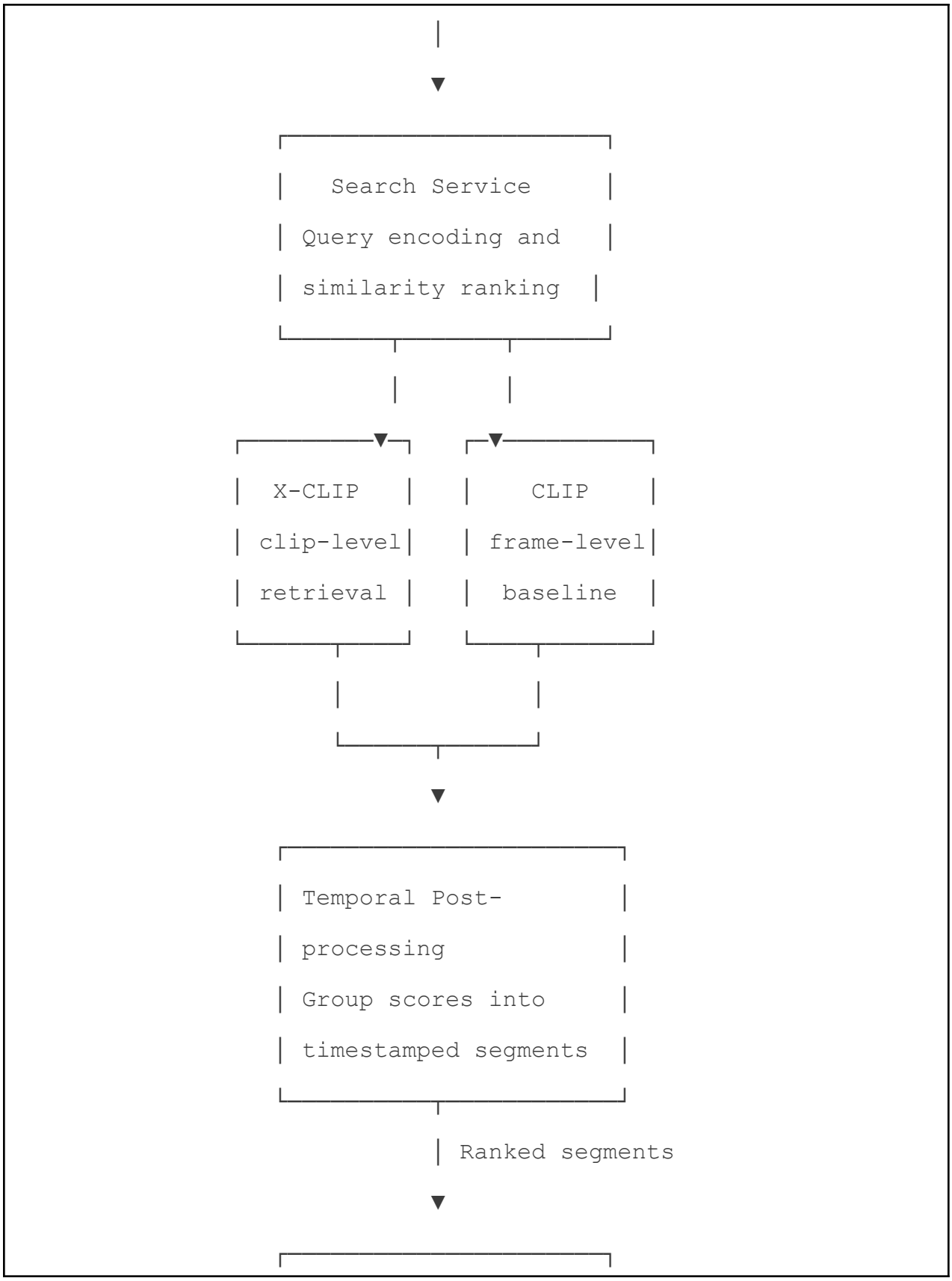
| Use case name | Brief Descriptions |
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| **Security Incident Investigation** | • Searching surveillance recordings for specific events using natural-language queries (e.g., "a person leaves a bag near an entrance").<br>• Retrieving relevant video segments with timestamps to help security personnel investigate reported incidents. |
| **Suspicious Activity Search** | • Finding potentially relevant activities, such as unauthorized entry, abandoned objects, or unusual movements.<br>• Searching across multiple cameras and recordings without manually reviewing the entire footage collection. |
| **Video Collection Management** | • Uploading surveillance videos and processing them asynchronously in the background.<br>• Browsing and filtering recordings by date, duration, filename, camera, or processing status. |
| **Targeted Video Retrieval** | • Searching within a selected subset of recordings, such as videos from a particular camera or time period. |

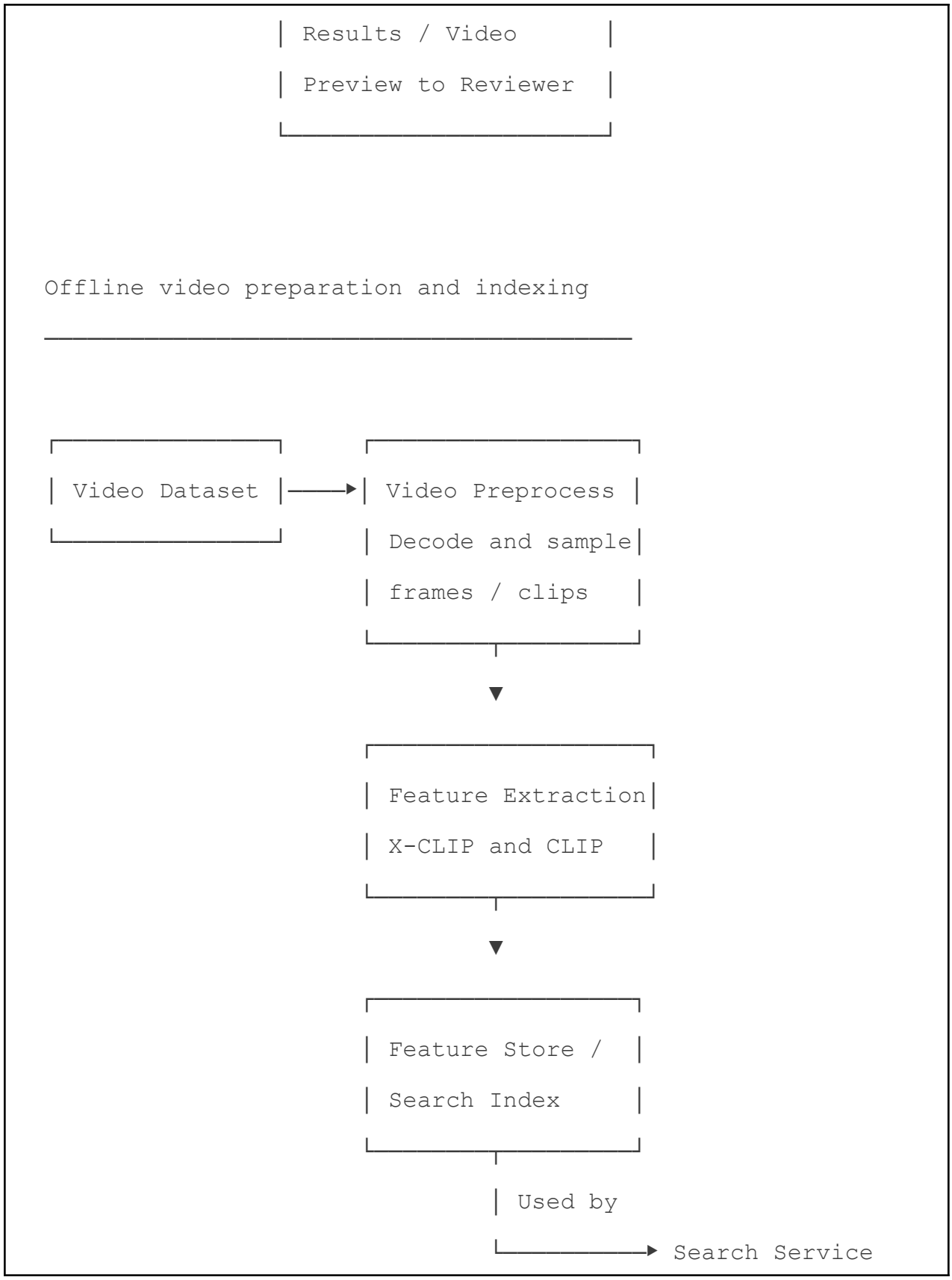
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| | • Ranking matching video segments according to their similarity to the natural-language query. |

| 4. Datasets to Be Employed / Created | |
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| **Public Datasets** | **Brief Descriptions** |
| **UCA (UCF-Crime Annotation)** | https://www.kaggle.com/datasets/vigneshwar472/ucaucf-crime-annotation-dataset<br><br>The dataset contains surveillance video recordings, containing event descriptions and temporal annotations. The dataset also contains category labels, but they won't be used. |

## 5. Project Architecture




Search Service
Query encoding and
similarity ranking
X-CLIP
clip-level
retrieval
CLIP
frame-level
baseline
Temporal Post-
processing
Group scores into
timestamped segments
Ranked segments


Results / Video
Preview to Reviewer
Offline video preparation and indexing
Video Dataset
Video Preprocess
Decode and sample
frames / clips
Feature Extraction
X-CLIP and CLIP
Feature Store /
Search Index
Used by
Search Service

## 6. API Resources (Web-Based Interfaces)

### REST Endpoints

- GET /videos - Fetch videos using optional filters such as upload date and processing status. Supports pagination and sorting.
- POST /videos - Upload a video for background processing. Returns 202 Accepted with a video identifier and processing-job identifier; the client can check progress without waiting for processing to finish.
- GET /jobs/{job_id} - Fetch the status of an upload/processing job, such as queued, processing, completed, or failed, including an error message if applicable.
- POST /search - Search video content using a natural-language query across the full indexed collection or a specified subset. Returns ranked candidate segments with video IDs, timestamps, and similarity scores.

## 7. Future Directions

The proposed system can be extended in several directions to improve retrieval accuracy, scalability, and practical usability.

### Real-Time and Near-Real-Time Video Search

Extend the current offline video-processing pipeline to support incoming video streams.

- Process video continuously or in short intervals as new footage becomes available.
- Retrieve potentially relevant segments without waiting for an entire recording to finish.
- Investigate the trade-off between sampling frequency, retrieval latency, and computational cost.

### Improved Temporal Event Localization

Improve the precision of the timestamps returned for natural-language queries.

- Explore temporal action localization models to identify more accurate event start and end times.
- Combine X-CLIP or CLIP similarity scores with temporal information to distinguish brief events from longer activities.
- Evaluate localization accuracy using temporal Intersection over Union (tIoU).

### Scalable Multi-Camera Search

Extend the system to handle larger video collections recorded by multiple cameras.

- Search across recordings from different cameras and time periods.
- Use vector databases or similarity-search libraries such as FAISS to accelerate retrieval over large feature collections.
- Investigate distributed feature extraction and indexing on HPC infrastructure.

## 8. Recommended Papers, Links & Resources

### Video-Language Models and Retrieval

- **"Expanding Language-Image Pretrained Models for General Video Recognition"** – Ni et al. (X-CLIP). https://arxiv.org/pdf/2208.02816
  Introduces X-CLIP, the primary video-language model used in this project, and explains how image-language representations can be extended to video.
- **"Learning Transferable Visual Models From Natural Language Supervision"** – Radford et al. (CLIP). https://proceedings.mlr.press/v139/radford21a
  Presents CLIP, which provides the image-text similarity baseline for comparison with X-CLIP.

### Surveillance Video Understanding and Datasets

- **"Towards Surveillance Video-and-Language Understanding: New Dataset Baselines and Challenges"** – Yuan et al. (CVPR 2024). https://openaccess.thecvf.com/content/CVPR2024/papers/Yuan_Towards_Surveillance_Video-and-Language_Understanding_New_Dataset_Baselines_and_Challenges_CVPR_2024_paper.pdf
  Introduces the UCA dataset, which extends UCF-Crime with fine-grained language descriptions and temporal annotations. It is particularly relevant to evaluating natural-language video retrieval.

### Video Retrieval and Temporal Localization

- **"Moment-DETR: End-to-End Video Moment Retrieval"** – Lei et al. https://arxiv.org/pdf/2107.09609
  Presents a method for locating relevant temporal moments in videos based on natural-language queries.